Control Group Consortium Analysis Report

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Key design decisions

We dissected the description of the research consortium into six sectors: finance, facility, equipment, consumables, staff and experiment-related scheduling, and participants. In total, the system consists of 29 entities. To make each sector distinct, we color-coded related entity tables using different colors.

When planning the queries for the web implementation, we narrowed the ERD down to 13 tables: institution, researcher, experiment, role, researcher_experiment, funding_source, experiment_funding_source, charge, charge_adjustment, facility, experiment_facility, equipment, and experiment_equipment. The queries support viewing all experiments, researchers, institutions, equipment, facilities, funding sources, charges, and charge adjustments, as well as the relationships between these entities through an interactive website.

On the website's home page, users can search for information related to keywords such as researchers, institutions, and experiments through a search bar. Below the search bar, users can add new experiments, researchers, institutions, charges, funding sources, and charge adjustments. The top navigation bar allows users to access all data related to the entities. When users select an entity, they can view its contents in the order they were added, sorted by unique ID. Users can filter data using dropdown menus available on different pages. On pages related to charge, users can also view the sum of different charges.

Lastly, a real-time experiment calendar is implemented on the website to visually display the duration and scheduling of experiments.

Assumptions

We had a few assumptions for our ERD listed below.

- One researcher belongs to only one institution
- A facility maps to only one institution
- facility_request references one specific facility
- Charges are tracked per event
- Every principal investigator is also a researcher
- facility_id represents equipment current location
- Reconfiguration and calibration are treated as maintenance events

- Information about where an equipment is located at is in the experiment_equipment table, just to reduce complexity, but it can be modeled out fully

Tradeoffs or limitations in your model

As mentioned above, we combined several aspects of the system, such as equipment information/location and maintenance events. While this reduces the model's complexity, it may overlook certain specific use cases. However, since we noted that these aspects can be modeled in greater detail, the system could be expanded further if needed. We will continue to refine this section as we work with the model and identify additional trade-offs or limitations that may arise.

In the web implementation, some limitations include the lack of date restrictions for records related to specific experiments, the presence of duplicated data with unique IDs, and the absence of an authentication system. Given the limited development time, we focused on covering the essential components of the organization and therefore had to simplify certain aspects of the system.

How ambiguities in the description were handled

There is an ambiguity surrounding the concept of an event in the description. To avoid overcomplicating the model, we treated an event not as a single entity, but as a combination of entities related to specific event categories such as maintenance or usage.

For the charge entity, specifically the responsible party attributes, we simplified the model by avoiding the creation of two separate entities related to either an institution or a researcher. Instead, we created two nullable foreign keys within the charge entity, allowing users to specify either a researcher or an institution as the responsible party.

In the web implementation, we further simplified some of these ambiguities to better accommodate the project workload.

How work was divided among team members

Our group faced a situation in which two group members were unresponsive to communication. This made working on the project more difficult, as a four-person project was being completed by only two members. We explained the situation to the professor, who allowed us to submit the project at a later date. Despite these challenges, we were able to draft the ERD, queries, web implementation, and analysis report while coordinating through text communication. Throughout the project, we shared equal responsibility in designing the entities and relationships, implementing the web application, and reviewing each other’s work. We followed the same collaborative approach when completing the analysis report.